

DOCX to PDF Converter

Two Python scripts for converting .docx files to PDF while preserving all formatting, styles, tables, images, and document elements.

Overview

This package provides two conversion approaches:

1. `docx_to_pdf.py` - Direct LibreOffice integration (recommended)
 - No additional Python dependencies
 - Works on all platforms with LibreOffice
 - Better control over conversion process
 - Detailed progress reporting
2. `simple_docx_to_pdf.py` - Using docx2pdf library
 - Requires `docx2pdf` pip package
 - Simpler code
 - Uses Microsoft Word on Windows, LibreOffice on Linux/Mac

Installation

Option 1: Using docx_to_pdf.py (Recommended)

Requirements:

- Python 3.6 or higher
- LibreOffice (free and open source)

Install LibreOffice:

```
# Ubuntu/Debian
sudo apt-get update
sudo apt-get install libreoffice

# Fedora/RHEL/CentOS
sudo dnf install libreoffice

# Arch Linux
```

```
sudo pacman -S libreoffice-fresh

# macOS (using Homebrew)
brew install --cask libreoffice

# Or download from: https://www.libreoffice.org/download
```

Windows:

Download and install from: <https://www.libreoffice.org/download>

No Python dependencies needed!

Option 2: Using `simple_docx_to_pdf.py`

Requirements:

- Python 3.6 or higher
- docx2pdf library
- LibreOffice (Linux/Mac) or Microsoft Word (Windows)

```
pip install docx2pdf
```

On Linux/Mac, also install LibreOffice (see above).

Usage

Method 1: `docx_to_pdf.py`

Convert a single file:

```
python docx_to_pdf.py document.docx
# Creates: document.pdf (in same directory)
```

Specify output filename:

```
python docx_to_pdf.py document.docx -o report.pdf
python docx_to_pdf.py document.docx --output /path/to/output.pdf
```

Convert all .docx files in a directory:

```
python docx_to_pdf.py /path/to/documents --batch
# Creates PDFs in same directory
```

Batch convert to different output directory:

```
python docx_to_pdf.py /path/to/documents --batch -o /path/to/output
```

Quiet mode (minimal output):

```
python docx_to_pdf.py document.docx -q
python docx_to_pdf.py document.docx --quiet
```

Complete options:

```
usage: docx_to_pdf.py [-h] [-o OUTPUT] [--batch] [-q] input

positional arguments:
  input                Input .docx file or directory (with --batch)

optional arguments:
  -h, --help            Show help message
  -o OUTPUT, --output OUTPUT
                        Output PDF file or directory (for batch mode)
  --batch               Convert all .docx files in the input directory
  -q, --quiet           Suppress verbose output
```

Method 2: simple_docx_to_pdf.py

Convert a single file:

```
python simple_docx_to_pdf.py document.docx
# Creates: document.pdf

python simple_docx_to_pdf.py document.docx output.pdf
# Creates: output.pdf
```

Convert all .docx files in a directory:

```
python simple_docx_to_pdf.py /path/to/documents --all
python simple_docx_to_pdf.py /path/to/documents /path/to/output --all
```

Features

Preserves All Formatting

Both scripts preserve:

- ✓ Text formatting (bold, italic, underline, colors)
- ✓ Font families and sizes
- ✓ Headings and styles
- ✓ Paragraphs and line spacing
- ✓ Tables with borders and shading
- ✓ Lists (bulleted and numbered)
- ✓ Images and graphics
- ✓ Headers and footers
- ✓ Page numbers
- ✓ Hyperlinks
- ✓ Table of contents
- ✓ Footnotes and endnotes
- ✓ Comments (as annotations in PDF)
- ✓ Tracked changes (rendered as final)
- ✓ Page layout and margins

Error Handling

- Validates input files exist and are .docx format
- Checks LibreOffice/Word installation
- Clear error messages
- Exit codes (0 = success, 1 = failure)
- Timeout protection (120 seconds per file)

Batch Processing

- Process entire directories of .docx files
- Summary statistics
- Individual file success/failure tracking

- Continues on error (doesn't stop batch on single failure)

Examples

Example 1: Convert technical documentation

```
# Single file with custom output
python docx_to_pdf.py "Technical_Guide.docx" -o "Technical_Guide_v1.0.pdf"
```

Example 2: Process curriculum documents

```
# Convert all curriculum files
python docx_to_pdf.py ./curriculum_docs --batch -o ./curriculum_pdfs
```

Example 3: Automated workflow

```
#!/bin/bash
# Convert all .docx files in current directory
for file in *.docx; do
    python docx_to_pdf.py "$file"
done
```

Example 4: Python script integration

```
from pathlib import Path
import subprocess

def convert_docx_to_pdf(docx_file):
    """Convert a docx file to PDF"""
    result = subprocess.run(
        ["python", "docx_to_pdf.py", str(docx_file)],
        capture_output=True,
        text=True
    )
    return result.returncode == 0

# Use it
```

```
if convert_docx_to_pdf("report.docx"):
    print("Conversion successful!")
else:
    print("Conversion failed!")
```

Troubleshooting

LibreOffice not found

Problem: "ERROR: LibreOffice not found!"

Solution:

1. Ensure LibreOffice is installed (see Installation section)
2. Check it's in your PATH: `which libreoffice` (Linux/Mac) or `where soffice.exe` (Windows)
3. Try running: `libreoffice --version`

Permission errors

Problem: "Permission denied" errors

Solution:

```
chmod +x docx_to_pdf.py
# Then run:
./docx_to_pdf.py document.docx
```

Timeout errors

Problem: "Conversion timed out"

Solution: Large files may take longer. The default timeout is 120 seconds. For very large files, you can modify the timeout in the script (search for `timeout=120`).

File format errors

Problem: "Input file must be .docx"

Solution: Only .docx format is supported. If you have legacy .doc files, convert them first:

```
# Using LibreOffice
libreoffice --headless --convert-to docx file.doc
```

Dependencies for docx2pdf

Problem: "ModuleNotFoundError: No module named 'docx2pdf'"

Solution:

```
pip install docx2pdf
# or
pip3 install docx2pdf
```

Platform-Specific Notes

Linux

- Works with any LibreOffice version (tested on 6.x, 7.x, 24.x)
- May need to install additional fonts for best results:

```
bash sudo apt-get install fonts-liberation fonts-crosextra-carlito
```

macOS

- LibreOffice from Homebrew works perfectly
- May need to grant Terminal permissions in System Preferences → Security & Privacy

Windows

- LibreOffice is recommended over Microsoft Word for automation
- If using docx2pdf with MS Word, ensure Word is properly licensed
- Path to LibreOffice: `C:\Program Files\LibreOffice\program\soffice.exe`

Performance

Conversion speeds (approximate):

- Small document (<10 pages): 1-3 seconds
- Medium document (10-50 pages): 3-10 seconds

- Large document (50-200 pages): 10-30 seconds
- Very large document (200+ pages): 30-60 seconds

Batch processing:

- 10 documents: ~30-60 seconds
- 100 documents: ~5-15 minutes

Technical Details

How it works

docx_to_pdf.py:

1. Locates LibreOffice installation
2. Invokes LibreOffice in headless mode
3. Uses LibreOffice's native conversion engine
4. Ensures all formatting is preserved through native rendering

simple_docx_to_pdf.py:

1. Uses docx2pdf library as wrapper
2. On Windows: Uses COM interface to Microsoft Word
3. On Linux/Mac: Uses LibreOffice via subprocess
4. Delegates actual conversion to the office application

Comparison

Feature	docx_to_pdf.py	simple_docx_to_pdf.py
Dependencies	None (just LibreOffice)	Requires pip install
Code complexity	Medium	Simple
Progress reporting	Detailed	Basic
Error handling	Comprehensive	Basic
Batch processing	Full-featured	Basic
Platform support	All	All (with caveats)

Feature	docx_to_pdf.py	simple_docx_to_pdf.py
Recommended for	Production use	Quick scripts

License

These scripts are provided as-is for converting .docx files to PDF format. LibreOffice is licensed under MPL 2.0.

Support

For issues with:

- **LibreOffice installation:** <https://www.libreoffice.org/get-help/>
- **docx2pdf library:** <https://github.com/AlJohri/docx2pdf>

Advanced Usage

Making the script executable

```
chmod +x docx_to_pdf.py
# Add shebang if not present (first line of script)
# Then run directly:
./docx_to_pdf.py document.docx
```

Adding to PATH for system-wide use

```
# Copy to a directory in PATH
sudo cp docx_to_pdf.py /usr/local/bin/docx2pdf
sudo chmod +x /usr/local/bin/docx2pdf

# Now use from anywhere:
docx2pdf document.docx
```

Integration with file managers

Nautilus (Ubuntu) - Right-click context menu:

Create `~/.local/share/nautilus/scripts/Convert to PDF` :

```
#!/bin/bash
python3 /path/to/docx_to_pdf.py "$1"
```

Make it executable:

```
chmod +x ~/.local/share/nautilus/scripts/Convert\ to\ PDF
```

Now you can right-click any .docx file and convert it!

FAQ

Q: Does this work with .doc (legacy) files?

A: No, only .docx. Convert .doc to .docx first using LibreOffice.

Q: Can I convert multiple files at once?

A: Yes! Use `--batch` mode to convert entire directories.

Q: Will this work without internet?

A: Yes! All conversion happens locally.

Q: Does it require Microsoft Office?

A: No! Works with free LibreOffice on all platforms.

Q: Is the output identical to the .docx?

A: Virtually identical. LibreOffice's PDF export is very accurate.

Q: Can I customize PDF settings (compression, quality)?

A: The scripts use LibreOffice's default settings. For custom settings, you'd need to modify the LibreOffice command parameters.

Q: Does it preserve hyperlinks?

A: Yes! All hyperlinks are preserved as clickable links in the PDF.

Q: What about password-protected .docx files?

A: These cannot be converted without first removing the password.

Enjoy your document conversions!  → 